CardioTwin

Causal forecasting of new atrial-fibrillation onset 2-10 minutes ahead

- CRP Digital Twin Challenge 2026 (Happiest Health) submission
- Team: Lazarus
- College / incubator: MGM's Jawaharlal Nehru Engineering College
- Team leader: Atharva Kulkarni; members Pranav Mirkar, Sumit Jadhav

Scope of this submission

- Decision every 30 s from the previous 10 min of ECG; target is a **new AF onset 120-600 s ahead**.
- **Research prototype** on retrospective public data. Not clinical, not a medical device, no automated treatment decisions.
- Honest result up front: the forecasting signal is weak. The rest of this deck shows how weak, and how we know.

Source: reports/ml-investigation/CHALLENGE_REQUIREMENTS.md.

Problem and India relevance

Early warning to support clinician review, not diagnosis.

Why this problem, and our scope

- The challenge asks for adverse-event prediction from EHR plus sensor data and a doctor-facing dashboard.
- We chose **one event: new AF onset**. Heart attack and stroke prediction are out of scope.
- Use case: flag a short window of higher risk so a clinician can review the ECG. Not diagnosis.
- Community AF estimates in India range from about **0.1% to 5%** depending on method and age, and national data are lacking.

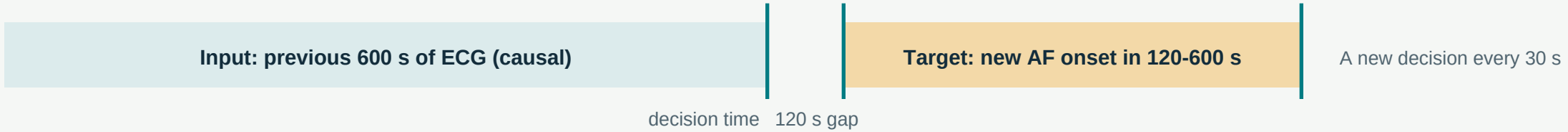
India context (cited; no causal claim)

- Rural Andhra Pradesh survey, 4,281 adults: AF **0.32%** (95% CI 0.17-0.54); 1.5% above age 60, 3.5% above 75. Saggu et al., Indian Heart J 2022.
- Small rural Gujarat screened sample (354 people aged 50+): 5.1% AF. Soni et al., JMIR Public Health Surveill 2016.
- India GBD 2019, **ischemic stroke only**: 535,700 incident cases and 271,200 deaths. Goyal et al., J Clin Med 2025.
- Kerala-AF Registry, 3,421 AF patients in 53 hospitals: NOAC use only about 5-6% in the 2,901-patient rate/rhythm cohort. Calvert et al., J Arrhythmia 2024.

Source: India: pmc.ncbi.nlm.nih.gov/articles/PMC9039683 (Saggu 2022); /PMC5083844 (Soni 2016); /PMC12609587 (Goyal 2025, GBD 2019); /PMC11317704 (Calvert 2024). Challenge: CHALLENGE_REQUIREMENTS.md.

Design principles

Causal inputs, patient-level splits, and every miss and false warning counted.

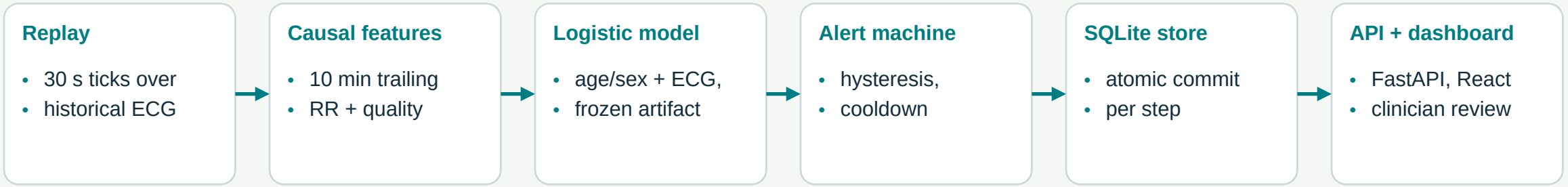


- **Causal:** only the previous 10 min of ECG; no future samples, annotations, onset or pair identifiers at runtime. The per-patient baseline is updated after each prediction.
- **Patient-level splits:** fit, early-stop, calibration and policy groups are disjoint. Model, calibration and alert policy were frozen before the held-out cohorts were opened.
- **Complete accounting:** report misses, false warnings per eligible hour, warning time and unassessable warnings, not just sensitivity.
- **Failed attempts stay on record:** gates were declared in advance and not loosened after results; held-out cohorts are not reused for tuning.

Source: README.md; reports/ml-investigation/report.md; frozen-challenge-fusion/config.json (history 600 s, gap 120 s, horizon 600 s, cadence 30 s).

Architecture

One shared alert and evaluation code path for runtime and offline scoring.



Offline only: labels, audit, training, complete-stream evaluation

- The labels module (future rhythm intervals) is **never imported** by runtime code.
- Alert and evaluation modules are shared, so the behaviour that is scored is the behaviour that is served.
- The default demo uses manufactured data and a synthetic-trained model. Opt-in `cardiotwin serve --research-model` runs the gate-failed frozen finalist in the twin under a red ‘RESEARCH MODEL ... FAILED its acceptance gates’ banner.

Source: README.md Architecture; docs/delivery/architecture.pdf.

Two data streams, one model

Static clinical fields plus dynamic ECG, fused in one logistic regression.

Static stream

- Code supports **16 static EHR-style fields**; the manufactured demo fills them.
- The real cohort provides **age and sex** only; nothing else is invented.

Fused predictor

- One logistic regression, **27 inputs**: age, sex, their availability flags and 23 ECG features. Development-only calibration, frozen before test.
- ECG-only reference is kept as a separate model.

Dynamic stream

- Causal ECG features: RR variability, beats, signal quality.
- Per-patient baseline is twin state shown to the clinician, **not a model input**.

Did fusion help? No benefit shown

- Development: ECG-only reference **7/241**, fused **6/241**; fusion showed no benefit.
- Held-out: fused 1/43 and 2/29; the reference issued no warnings (policy threshold 1.0).

- Baseline deviation is computed for display; no baseline feature is among the 27 model inputs.

Source: CHALLENGE_REQUIREMENTS.md; README.md; frozen-challenge-fusion/feature_schema.json (27 inputs); src/cardiotwin/features.py (CLINICAL_FIELDS); report.md.

Data

Public, retrospective, anonymized recordings only.

Cohort	Groups	Eligible AF onsets	Role
IRIDIA-AF development	121 patient groups (132 recordings)	241	Grouped 3-fold development: fit, stop, calibration, policy
IRIDIA-AF primary held-out	31 patient groups	43	Opened once, after the model was frozen
LTAF external	84 recording groups	29	External transfer; patient independence unverified

- IRIDIA-AF in total: **167 recordings, 152 patients, 6,690.476 monitoring hours**; 284 eligible events across 143 patients (241 + 43).
- Development monitored time: 5,243.98 h; 233 of 241 events have at least one available decision.
- Missing clinical fields stay missing (LTAF has no age/sex). Synthetic fixtures validate software only and are not evidence.
- Raw data and credentials stay outside the public repository.

Source: README.md Evidence-driven investigation; report.md; docs/video-script.md results card. IRIDIA-AF: Gilon et al. 2023, CC BY 4.0. LTAF: Petrutiu et al. 2007.

Evaluation method

Whole recordings replayed in time order, nothing filtered out afterwards.

How we score

- Recordings replayed chronologically; quality-rejected stretches and missed events stay in.
- One warning matches at most one event; false warnings per **eligible hour**.
- Unknown-outcome and unassessable warnings (during AF, post-AF washout) are reported separately.
- Also reported: warning occupancy, coverage, calibration, lead time, patient-group bootstrap intervals.

Our own criteria (not the organizers')

- Research assumption: at most **0.25 assessed false warnings per eligible hour**.
- Research assumption: **50% event sensitivity** as a provisional engineering gate.
- The organizers published no accuracy threshold or rubric; these are ours and chosen before the held-out look.
- Gate result: sensitivity 2-7% against 50%, so the gate **fails**.

Source: README.md Model and evaluation protocol; CHALLENGE_REQUIREMENTS.md (organizers published no accuracy threshold).

Dashboard demo (manufactured data)

Software demonstration of the doctor-facing view; synthetic data is not evidence.

What the UI shows

- ECG, trends, per-patient baseline
- Signal quality and abstention
- Risk, warning history
- Clinician review of each warning

What it is not

- Default demo data are manufactured, not real patients.
- Default model is synthetic-trained; the real finalist runs only via opt-in `--research-model`, red FAILED banner.
- No hospital login, PHI governance or wearable validation.

Causality and speed checks

- Replay step latency on real prefixes: median **25 ms**, p95 **27 ms** (72 steps).
- Changing only future ECG or annotations left earlier outputs identical.

- Local proof, **in-sample**: frozen finalist on one IRIDIA development record gave forecasts at 3600, 7200 and 10800 s (risks 0.0092, 0.0079, 0.0080), 0 warnings, about 26-29 ms per step. This proves the model runs in the twin, not forecasting quality.
- Tests: full suite in this public repo on 17 Oct 2026: 841 passed, 83 skipped (private receipts or optional PyTorch), 0 failed.

Source: reports/ml-investigation/frozen-challenge-replay.json (72 steps; passed: true); src/cardiotwin/cli.py and tests/test_research_serving.py (--research-model); research-model run was a local check, no saved receipt; full suite 841 passed, 83 skipped, 0 failed.

Results: weak

Frozen age/sex + ECG logistic finalist; events detected out of eligible events.

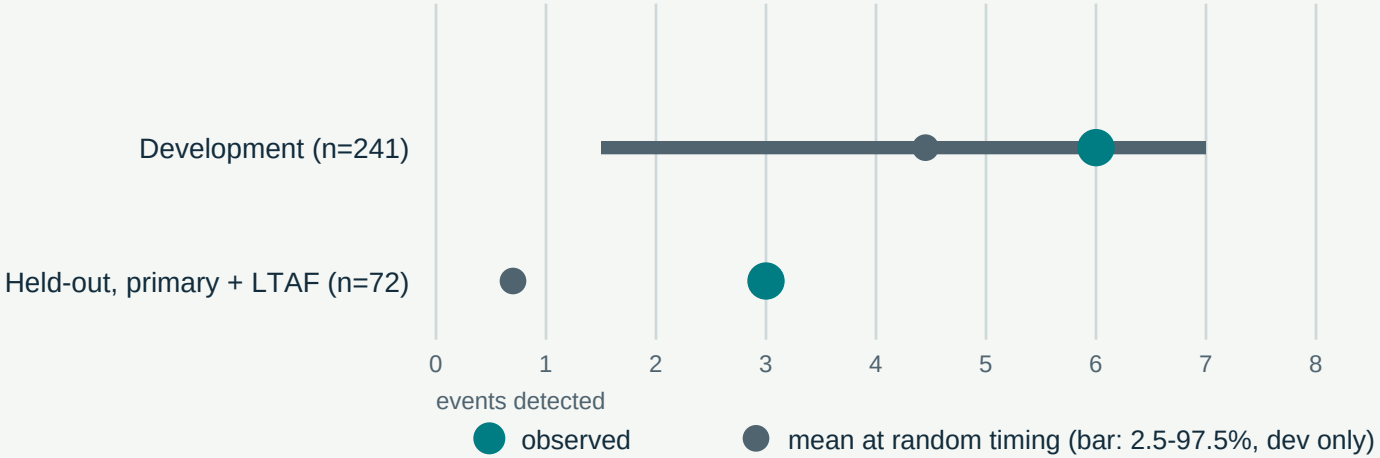
Cohort	Detected	Sensitivity [95% CI]	False warnings / eligible h	Warning starts	Occupancy	Window AUROC
Development (121 groups)	6 / 241	2.49% [0.44, 4.96]	0.171	761	40.8%	0.45 pooled
Primary held-out (31)	1 / 43	2.33% [0.00, 7.89]	0.085	113	1.7%	0.611 [0.542, 0.681]
LTAF external (29 events)	2 / 29	6.90% [0.00, 16.67]	0.008	398	6.2%	0.631 [0.513, 0.761]
ECG-only reference	dev 7/241; primary 0/43; LTAF 0/29	reference primary and LTAF issued no warnings	-	-	-	-

- Sensitivity is **2-7%** against our 50% target; assessed false warnings are under our 0.25/h budget, but sensitivity is far below target.
- Dev AUROC by fold 0.40 / 0.49 / 0.57: no reliable ranking. Held-out AUROC 0.61-0.63 is modest.
- Probabilities add nothing over base rate in development: Brier 0.008384 vs 0.008358 for a constant-prevalence predictor.

Source: Primary/LTAF: saved hash-bound evaluation.json files. Development and AUROC: reports/ml-investigation/chance-analysis/{dev_summary,brier_results,heldout_results}.json; report.md.

Against chance

Same alert policy re-applied to risk shifted to random times inside each recording.



- **Development:** 6 observed vs about 4.5 expected; chance of at least 6 is 0.33. Indistinguishable from random timing.
- **Held-out:** 3 of 72 vs about 0.7 expected (uncorrected p about 0.035). Three events, intervals include zero, post-hoc: **no skill claim**.

LTAF low false-warning rate is not skill

- 392 of 398 warning starts fall in unassessable time (222 during AF, 77 washout, 93 unknown).
- Random placement of the same warning time would give about 0.10 assessed false warnings/h vs our 0.008.

One exception, not our finalist

- RR-TCN16 patient/class: 7/241 vs 1.5 expected, fold 0 only, 1 of 8 arms. Bonferroni-adjusted p about 0.13.

Source: reports/ml-investigation/chance-analysis/dev_summary.json (60 draws), heldout_results.json (200 draws); report.md LTAF burden.

What failed

Many model families, all within 1-10 of 241 development events.

Attempt	Dev events	Note
Age/sex + ECG logistic (frozen finalist)	6 / 241	ECG-only reference 7 / 241: clinical fields gave no benefit
Gradient boosting, per-patient baseline, ectopy-like motifs	1-10 / 241	Classical sweeps, all within chance range
RR-TCN16 (RR-interval CNN), two arms	4 / 241 and 7 / 241	Only 7/241 beat random timing, fold 0 only
Waveform CNN with causal TCN head	4 / 241	33% warning occupancy; window AUROC about 0.458
Precursor ranking	5 / 241	Gains in fold 1 only
ECGFounder foundation-model attempt	no results	0 outcome fits; last Kaggle job failed in pip setup before reading any data

Calibration defect found, fixed in code, not used to re-select

- Fold-0 calibrators learned **negative slopes** and reversed ranking (AUROC 0.573 raw to 0.427). New slope guard in train.py::_calibrate rejects such fits; test added.
- Frozen finalist and reference had positive slopes, so held-out results are unaffected. No sweeps after this: 241 dev events invite winner's curse.

Source: README.md; reports/ml-investigation/{report.md, classical-calibration-diagnostic.json, chance-analysis/*.json}; src/cardiotwin/train.py; tests/test_calibration_slope.py.

Limitations and validation plan

Public retrospective data; the next step needs new data and pre-declared gates.

Limitations

- Public retrospective data; no Indian-population evidence; not a medical device.
- Primary and LTAF cohorts are used up: no tuning on them.
- Development numbers come from choosing among many candidates and are exploratory.
- LTAF groups are recording IDs; patient independence unverified.
- Target is AF onset only, not heart attack, stroke or treatment advice.

Future validation plan

- Ethically approved Indian clinical partner and data (none secured yet).
- Fresh cohort with model, metrics and gates declared before looking at it.
- Device and patient-group calibration, then prospective silent (non-interventional) evaluation.
- No care use before that evidence exists.

- Repository: https://github.com/sumitjadhav1703/lazarus_ECG_traces
- Demo video: **[UNLISTED YOUTUBE URL - to be supplied]**

Source: Limits: README.md, reports/ml-investigation/report.md. Attribution: Gilon et al. 2023, IRIDIA-AF v1.0.1 (zenodo.org/records/8405941, CC BY 4.0); Petrutiu et al. 2007, LTAF v1.0.0 (physionet.org/content/ltafdb/1.0.0/); full text in DATA_LICENSE.md.